Time-Based Crop Yield Forecasting with a Multilayer Perceptron

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Course: Machine Learning

1. Performance

1.1 Evaluation Metrics

The performance of the model was evaluated using distance-based and correlation-based metrics. The distance metrics reflect the distance between the prediction and the actual yield values and the correlation metrics reflect the way the prediction follows the true trend. Together, they evaluate both error magnitude and the model's ability to explain yield variance. The three metrics used are:

1. **Root Mean Squared Error (RMSE):** It measures the average squared prediction error, penalizing larger errors more heavily. This metric is the scale of total error of the model in kg/ha, the same unit as of the target variable.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}$$

2. **Mean Absolute Error (MAE):** It measures the average absolute difference between predicted and actual values, offering a more interpretable error estimate. Unlike RMSE, it is not that sensitive to outliers.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i|$$

3. **R^2 Score:** It captures how well the model explains the variance in the actual yield values. A higher R^2 value indicates that the predictions follow the overall trend of the true data more closely.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{1=1}^{n}(y_i - \bar{y}_i)^2}$$

Where n refers to the number of data instances in the test set, $\hat{y}_i$ represents the predicted yield, y_i is the actual yield, and $\bar{y}_i$ is the mean of the actual yield values for instances *i*. RMSE and MAE quantify prediction error magnitude, while R^2 measures how well the model captures the true data's underlying structure.

1.2 Test Set Results

After training and validation on past data, the model was evaluated on the test set having 2021 instances to forecast 2022 yield. Table 1 shows strong performance, with the model explaining 91% of yield variance. The low RMSE and MAE values further confirm its strong predictive performance, with an average error of 1,597 kg/ha, reasonable across various crops and regions.

Metric	Value
RMSE (kg/ha)	4455.12
MAE (kg/ha)	1597.33
R^2 (kg/ha)	0.91

Table 1. Test Set Performance

1.3 Predicted vs Actual Plot

Figure 1 shows a scatter plot of predicted vs. actual yield values for 2022, where each point is a unique combination of country-crop. The red line indicates perfect predictions, and most points are aligning close, which means high model accuracy. Some deviations in high yield ranges may be caused by extreme environmental conditions, or crops that are not well represented. In spite of these, the overall congruence underpins the high R^2 score.

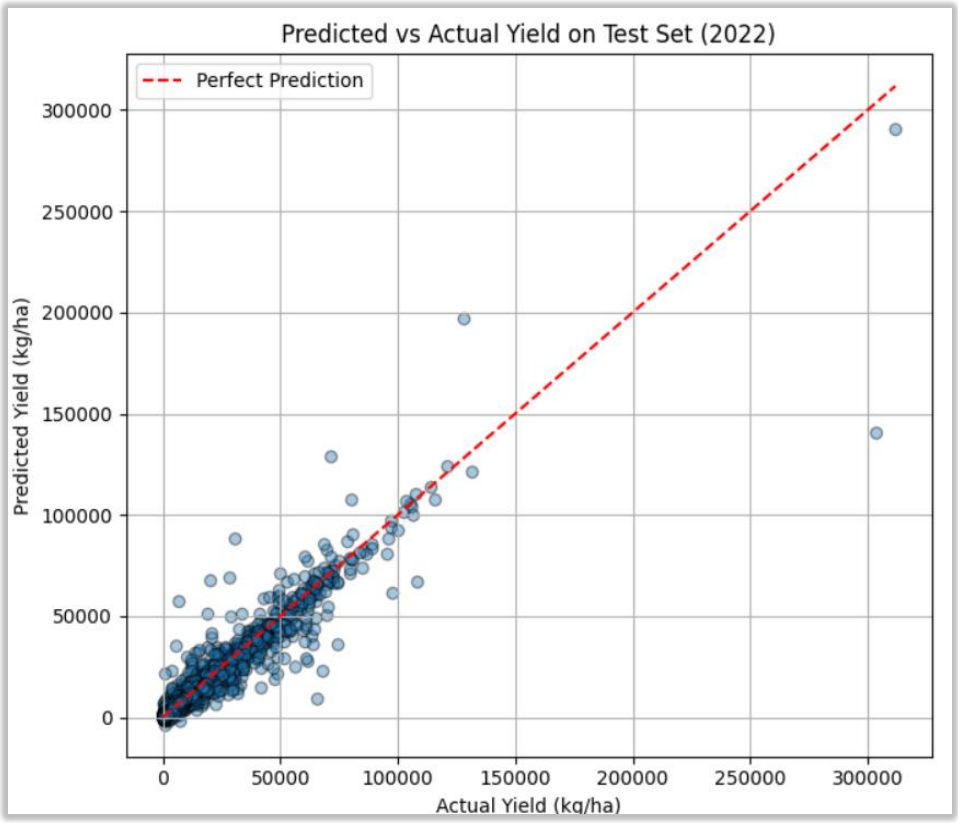

Figure 1. Predicted vs Actual Yield for the 2022 (kg/ha)

1.4 Data Splitting Strategy

A time-based split simulated real-world forecasting as future outcomes were predicted based on past data. Unlike the random partitioning methods like Scikit-learn's train_test_split that can combine past and future instances leading to data leakage, with this method chronological order was maintained. As yield is influenced normally by such dynamic factors as technology, climate change, and farming practices, time consistency was essential. Causal realism and avoidance of future data influencing the predictions was achieved by training on earlier years and testing on later ones.

After all preprocessing, the final dataset consisted of 57,447 instances. The data was then divided into the following sets:

- Training subset: data from 2010 to 2019 (predicting yield for 2011 – 2020), comprising 47,079 instances.
- Validation set: 2020 data (predicting yield in 2021), comprising 5,178 instances.

- Test set: 2021 data (predicting yield in 2022), comprising 5,190 instances.

This corresponds approximately to an 80:10:10 split for training, validation, and test sets respectively. This split design emulates a real-world forecasting environment and guarantees that the model is only subjected to historical data when training. The test set reflects a truly unseen future, supporting a fair and unbiased evaluation of the model's predictive capability.

1.5 Summary

The final model outperformed both the R^2 and error score on the 2022 test set, validating both accuracy and generalisability. The scatter plot demonstrates high correlation of predicted and actual values for different crops and countries. Time aware split and strict separation of future data ensured best practices of forecasting were followed.

2. Model

The model MLP (Multilayer Perceptron), a feedforward neural network capable of capturing complicated and non-linear relations between high dimensional inputs and continuous outputs were used. It was the final stage in a pipeline that included preprocessing, feature encoding, scaling, and supervised training with validation-based early stopping.

2.1 Architecture and Computation

The model architecture consists of an input layer, two fully connected hidden layers and one output node. The input layer processes scaled numerical and one-hot encoded categorical features. Hidden layers have 128 and 64 units, and both use ReLU (Rectified Linear Unit) activation: $\sigma(z) = \max(0, z)$. ReLU introduces non-linearity without vanishing gradient issues. The final output layer has a single unit with linear activation, ideal for predicting continuous values. The feedforward process is explained as:

$$h_1 = \sigma(w_1 \cdot x + b_1), h_2 = \sigma(w_2 \cdot h_1 + b_2), \hat{y} = w_3 \cdot h_2 + b_3$$

where x is the input feature vector, $\hat{y}$ is the predicted yield, and W, b represents weights and biases of each layer.

2.2 Hyperparameters and Overfitting Prevention Strategies

The model was trained using stochastic gradient descent with the Adam optimizer and a learning rate of 0.0005, selected through empirical testing for optimal balance between convergence speed and stability. The loss function was Mean Squared Error (MSE) which was perfect for regression problems and heavily penalised larger deviations. The MSE loss is given as:

$$L = \frac{1}{2}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

Where $\hat{y}_i$ is the predicted yield, y_i is the actual yield for the instances *i*.

Batch sizes of 16, 32 and 64 were investigated where 32 was selected to avoid overfitting. The model was first trained for 200 epochs. In order to avoid overfitting, early stopping was used during training to determine the point at which the generalisation performance of the model no longer increased. Following 200 epochs of training with validation loss being recorded at every epoch, the training and validation loss curves were

plotted. From the Figure 2, it was noted that validation loss converged at epoch 171. After this point, no substantial increase in performance was observed. Therefore, the model was re-trained with this early stopping point as the final number of epochs that helped to capture the optimal level of generalisation and reduced the risk of overfitting. A validation set was used to monitor this model performance and determine the optimal stopping point during training.

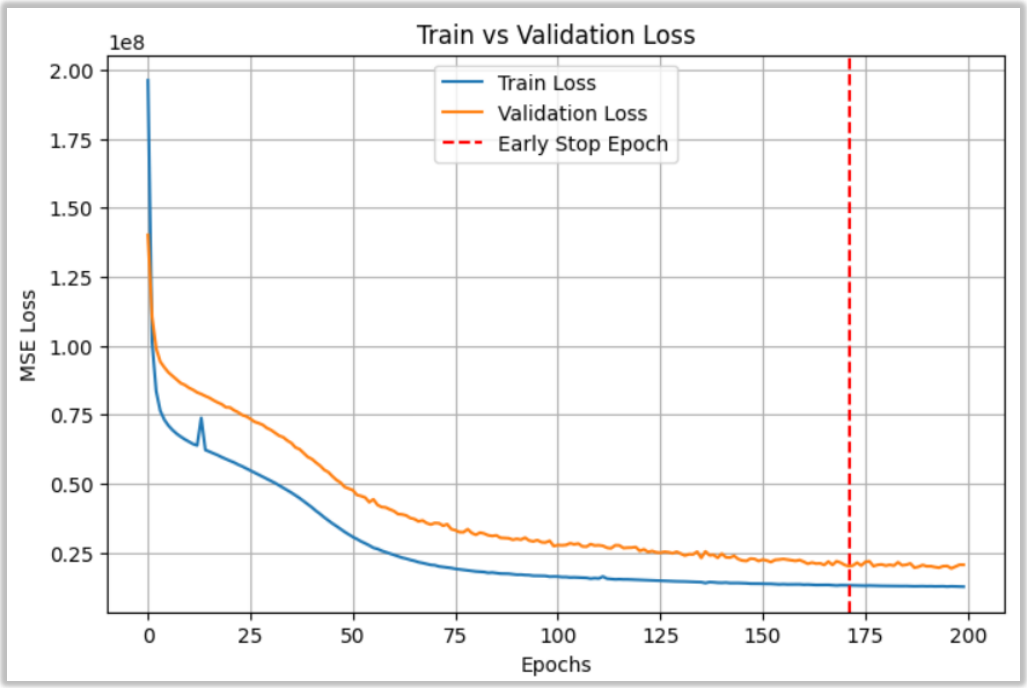

Figure 2. Training and Validation Loss over Epochs

To improve generalisation, a time-based validation split was used to test the model on future data rather than a random sample, minimising information leakage. All numerical features were normalised to zero mean and unit variance for standardisation across all data and better optimisation dynamics. The architecture of the model was made moderate in order to achieve a balance between learning capacity and overparameterization. Hidden layers provided enough representational power without encouraging overfitting. The training set included 47,079 varied instances from countries, crop classes, and environment, thus making the training statistically strong and implementable in real world prediction.

3. Features & Labels

It describes how each feature was extracted, transformed, and chosen and describes why is it included or excluded according to structure, relevance, and predictive value.

3.1 Output Data

The model's goal was to predict annual crop yield per hectare. This value was taken from the *'Yield_and_Production_data.csv'* dataset, where the target variable was initially stored under the *'Value'* column.

The selected columns from this dataset were *'Country'*, *'Item'*, *'Year'*, and *'Value'*. These were renamed to *'country'*, *'item'*, *'year'*, and *'yield_kg_per_ha'* for consistency and readability.

Thus, the target label used for training and evaluation was *'yield_kg_per_ha'*, a continuous numerical variable representing crop yield in kilograms per hectare.

3.2 Input Data (Features)

The input features were extracted from a range of datasets, which included both environmental data and contextual information. After preprocessing, the model used a total of 296 input features.

3.2.1 Environmental Features (Numerical)

Environmental features included rainfall, snowfall, soil moisture and temperature at various depths, transpiration, soil evaporation, terrestrial water storage, canopy water content, and land cover composition. The following datasets contributed to the environmental features:

- *'Rainf_tavg_data'*
- *'Snowf_tavg_data'*
- *'SoilMoi0_10cm_inst_data','SoilMoi10_40cm_inst_data', SoilMoi100_200cm_inst_data'*
- *'SoilTMP0_10cm_inst_data','SoilTMP10_40cm_inst_data', 'SoilTMP40_100cm_inst_data','SoilTMP100_200cm_inst_data'*
- *'TVeg_tavg_data'*
- *'ESoil_tavg_data'*
- *'TWS_inst_data'*
- *'CanopInt_inst_data'*
- *'Land_cover_percent_data'*

Each environmental dataset provided monthly data (12 months), which was aggregated into annual averages. This resulted in one numerical value per feature, per year, per country, preserving seasonal trends while reducing dimensionality. The *'Land_cover_percent_data'* dataset was already annual and used directly without aggregation. All datasets were merged using inner join on *'latitude'*, *'longitude'*, and *'year'*.

A *'country_latitude_longitude_area_lookup'* dataset was used to align coordinates with country names. Only *'country', 'latitude', and 'longitude'* from this dataset were retained; the *'radius'* field and other administrative columns were excluded, as spatial interpolation was not required.

3.2.2 Categorical Features

The columns *'country' and 'item'* were used to capture regional and crop-specific differences. These were one-hot encoded into 268 binary columns. This change allowed the model to distinguish meaningful differences between locations and crop types without imposing ordinal relationships.

The *'year'* column was excluded from the input features. Though, the year column was needed for time-based alignment and splitting, the inclusion of it into training could introduce artificial time-based signals, which could decrease the model's ability to generalize.

3.3 Feature Retention and Justification

After all preprocessing steps and merging, the final dataframe contained **57,447 records** and **298 columns**, comprising:

- **297 input features**, including:
  - 9 aggregated and annualised environmental features ('rainfall_avg', 'snowfall_avg', 'soil_evap_avg', 'canopy_water_avg', 'transpiration_avg', 'soil_moisture_0_10_avg', 'soil_moisture_10_40_avg', soil_moisture_100_200_avg', 'soil_temp_0_10_avg')
  - 17 land cover percentage features
  - 2 Geographic features ('latitude' and 'longitude')
  - 268 one-hot encoded categorical columns (‘country’ and ‘item’)
  - 1 year column (not used for training but retained for include in final csv file)
- **1 target variable**: 'yield_kg_per_ha'

While a correlation matrix showed weak linear associations with yield, this was expected, as yield is influenced by non-linear, multi-feature interactions. The MLP architecture was well-suited to capture these patterns, so all variables were retained. Histograms with KDE plots were also used to visualise feature distributions. While several variables, such as 'snowfall_avg' and certain land cover classes, were highly skewed, they were not removed. These values reflected meaningful conditions (e.g., absence of snowfall in tropical regions) rather than errors or noise. All numerical features were standardised before training.

Columns such as *'Domain', 'Flag', 'Item Code (CPC)', and 'Element'* from the original yield dataset were also excluded. These represented metadata, duplicate encodings, or quality indicators not relevant for prediction and would have added unnecessary noise.

In summary, all meaningful and relevant features were retained, with clear justification for each inclusion or exclusion. This allowed the model to fully leverage the rich spatial, temporal, and categorical variation present in the dataset.

4. Preprocessing

These steps included data cleaning, feature aggregation, merging, data splitting, encoding, scaling, and output file preparation. Each performed systematically to align with the model’s requirements and the real-world forecasting setup. I loaded data from multiple CSV files using the pd.read_csv() function.

4.1 Data preparation

Annual averages were computed for each environmental variable, yielding one value per region per year as discussed in section 3. Rows with missing values (NaN) were removed from 'SoilMoi100_200cm_inst_data.csv' and 'TWS_inst_data.csv' (14,946 rows) and 6 rows from 'country_latitude_longitude_area_lookup.csv,' ensuring data integrity across merged sources. Zero values were retained, as they represent valid environmental conditions (e.g., no snowfall in tropical regions or 0% for absent land cover classes). No duplicate rows were

found. Although many outliers were detected, they were preserved, as they provide meaningful insights, similar to zero values. To isolate yield data, the dataset was filtered to include only rows where 'Element' = 'Yield.

4.2 Merging

Firstly, I merged all the environmental datasets using inner join on the key features latitude, longitude and year. Secondly, to support spatial alignment between datasets with differing structures (e.g., environmental grids vs. country-level yield records), a K-Nearest Neighbours (KNN) approach was used. Using the latitude and longitude coordinates from the *'country_latitude_longitude_area_lookup.csv'*, each country was mapped to the closest location in the environmental datasets based on Euclidean distance. This ensured that each country's annual yield was matched with the most geographically relevant environmental features. The KNN algorithm was implemented using Scikit-learn's KNeighborsClassifier with "metric = 'euclidean'" and k = 1. The choice of k = 1 ensured that each country was mapped to its single nearest environmental grid point, avoiding any averaging across potentially distant or irrelevant locations. This approach provided a precise and consistent way to align country-level yield records with spatially granular environmental features. Finally, I merged these with *'Yield_and_Production_data.csv'* data using inner join on 'country' and 'year'.

4.3 Time Shift: Aligning Features and Labels

The forecasting task aimed to predict next year's crop yield using current year's environmental data. For example, features from 2021 should predict the yield in 2022, mirroring real-world forecasting, where only past and present data inform future predictions. However, the fully merged environmental data covers the years 2010 to 2022, while the original yield data covers 2010 to 2022 as well. However, to enable forecasting, the *'year'* column in the *'Yield_and_Production_data'* dataset was shifted backward by 1. After this shift, the yield data spans 2009 to 2021. When the two datasets were merged using an inner join on *'year'* and *'country'*, the overlapping year range became 2010 to 2021. This means that each row in the merged dataset uses environmental features from year t and yield from year t+1. The table 2 shows how environmental features from year t are aligned with crop yield from year t+1. The feature values shown are illustrative for understanding purpose.

Year (Used in Model)	Input features	yield_kg_per_ha (Target)
2010	….	Yield in 2011
2011	….	Yield in 2012
….	….	….
2020	….	Yield in 2021
2021	….	Yield in 2022

Table 2. Alignment of Features and Yield After Time Shift

Thus, the model was trained using data where *'year'* ranged from 2010 to 2019 meaning it learned to predict yields from 2011 to 2020 using environmental features from 2010 to 2019.

It was validated on the year 2020 (predicting 2021 yield), and finally tested using 2021 environmental data to forecast the yield for 2022 a year it had never seen during training.

4.4 Encoding and Scaling

Categorical variables ('country' and 'item') were one-hot encoded, producing 268 binary columns. All numerical features were standardised using StandardScaler, transforming them to zero mean and unit variance before training the MLP model. This step was essential for ensuring stable and efficient training of the neural network, as features on different scales could otherwise dominate the learning process.

4.5 Data Splitting

The data was split chronologically to preserve the forecasting structure as discussed in section 1. The model was trained on data from 2010 to 2019, validated on 2020, and tested on 2021 corresponding to yield predictions for 2011 – 2020, 2021, and 2022, respectively. This non-random, time-aware split prevented information, where predictions are always made about the future using only past data.

4.6 Result Dataset Preparation

After prediction, the features were reverted to their original scale and stored in the output dataset, which includes the columns: *'year,' 'country,' 'item,' 'actual_yield,' 'predicted_yield,' and 'percent_diff.'* Predicted values were rounded to one decimal place. The *'percent_diff'* column represents the percentage difference between actual and predicted yields, with negative values indicating underestimation and positive values reflecting overestimation. The model's results were saved as a CSV file.